

Requirements- based testing of neural networks.

A critical review of RBT4DNN, its empirical evidence, and two focused additional analyses.

From a requirement to credible evidence.

01	02	03	04	05	06
Problem statement	Background	Current approach	Methodology	Current results	Critical review
Why test by requirement?	Terms and assumptions	RBT4DNN overview	Data, labels, generator adaptation	RQ1-RQ6	Additional analyses + future scope

WHY REQUIREMENTS?

HOW DOES RBT4DNN WORK?

HOW STRONG IS THE EVIDENCE?

High average accuracy does not answer a specific requirement.

CONVENTIONAL TESTING Known behavior A specification maps an input condition to an expected output. Test oracles can check the result directly.	DNN TESTING Learned behavior The model is inferred from examples. Global accuracy summarizes a distribution, not every meaningful scenario.	THE MISSING LINK Semantic scenarios Developers need evidence for statements such as “a very thick 7 must still be classified as 7.”
GLOBAL METRIC How often is the complete test set correct?	→ SEMANTIC SLICE Which rare inputs does the score contain?	→ REQUIREMENT VERDICT Does the model satisfy this specific contract?

[1] §1-§2.5 · “model under test” means the evaluated neural network

Existing generators rarely start from the requirement itself.

Neuron coverage, image transformations, and search-based generators can produce many tests—without proving that those tests represent the scenario a stakeholder cares about.

INPUT	Requirement	Structured natural language defines a semantic precondition and an expected outcome.
METHOD	RBT4DNN	Fine-tune a generator on images that satisfy that precondition.
OUTPUT	Targeted tests	Run the learned component and evaluate the requirement-specific oracle.

[1] §1, §2.5 and §3 - authors claim the first requirement-driven DNN test-input generator

The study spans four domains, four output regimes, and 25 requirements.

MNIST · 7 REQUIREMENTS MNIST artifact samples Source-linked in interactive deck Digits Measured height, thickness, and lean. A controlled starting point. Output: digit class	CELEBA-HQ · 7 REQUIREMENTS CelebA-HQ artifact samples Source-linked in interactive deck Faces Ambiguous attributes labeled by visual question answering or humans. Output: eyeglasses decision	SYNTHETIC DRIVING (SGSM) · 7 SGSM artifact samples Source-linked in interactive deck Driving Scene relations linked to steering and acceleration constraints. Output: numeric control	IMAGENET · 4 REQUIREMENTS ImageNet artifact samples Source-linked in interactive deck Animals Morphology linked to allowed classes in the WordNet taxonomy. Output: class set
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[1] Table 2 and §4.1.1-§4.1.3 · samples linked from the pinned public artifact

MNIST M3 artifact samples
Source-linked in interactive deck

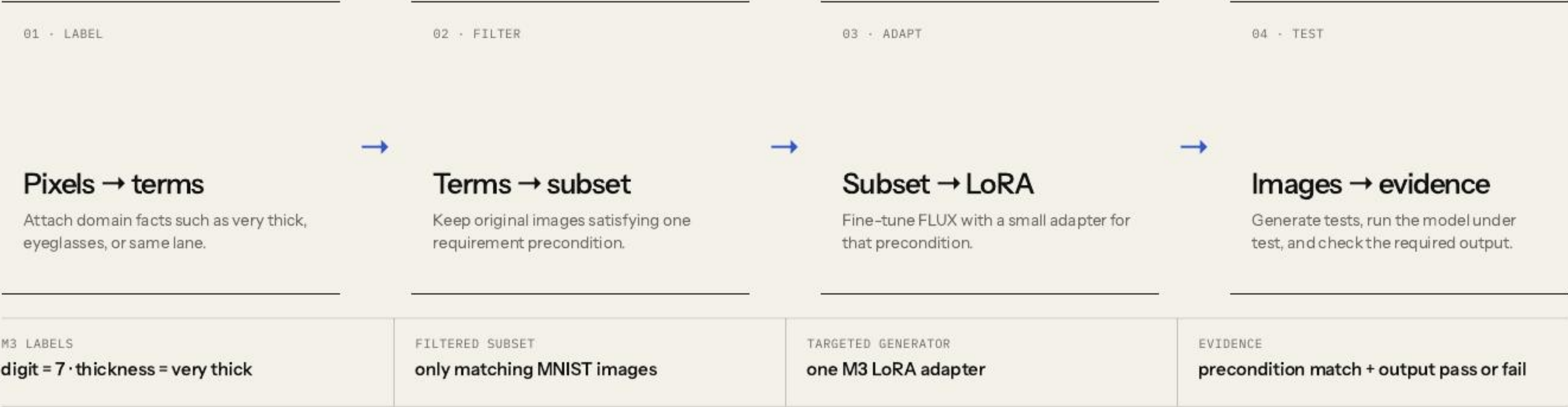
A requirement defines both relevance and correctness.

If the digit is a 7 and is very thick, then the model shall label it as 7.

PRECONDITION	POSTCONDITION
Which images count as relevant tests?	Which model outputs count as acceptable?

[1] Table 2, M3 · a finite passing suite cannot prove the universal requirement

RBT4DNN is a four-stage semantic pipeline.



[1] Fig. 2 and §3 · symbols are translated into an explicit input/output contract

Glossary terms make visual requirements operational.

GLOSSARY TERM

A named visual property that can be assigned to an image: “very thick,” “same lane,” or “eyeglasses.”

01 · DEFINE

Authors specify the glossary

Requirement concepts are manually turned into a finite vocabulary of visual properties.

TERM CREATION IS NOT AUTOMATIC

02 · LABEL ORIGINAL DATA

Labels come from domain-specific signals

MNIST measurements · SGSM scene graphs · MiniCPM visual Q&A for CelebA-HQ and ImageNet.

CELEBA ALSO HAS HUMAN LABELS

03 · USE THE LABELS

Filter first; check later

Labels select the LoRA training subset. Separate binary classifiers then check glossary terms on generated images.

PRECONDITION MATCH · RQ1 / RQ3

Boundary: separate training labels and evaluation classifiers reduce direct reuse, but neither constitutes independent semantic ground truth.

[1] §3.2 and §4.1.6 · mean glossary-classifier accuracy 94.5%; C4 is 70.8%

One filtered precondition subset trains one targeted LoRA.



BENEFIT	Generate many new candidates inside a rare requirement region
EVIDENCE SOUGHT	Relevance, distributional plausibility, and semantic variation
NOT ESTABLISHED	Non-memorization or systematic coverage within that region

[1] §3.3 and §4.1.4 - LoRA equation simplified; latent-space coverage remains future work

The baselines are not interchangeable.

PRIMARY METHOD Requirement-specific LoRA Filtered precondition subset. The central RBT4DNN configuration.	MNIST CONSISTENCY ABLATION All-data LoRA Same domain and prompting, but the adapter sees the complete dataset.
DISTRIBUTIONAL QUALITY Prompt-only FLUX No domain adaptation. Used for distributional quality comparisons.	MNIST TESTING BASELINES DeepHyperion + transformations Established test generators that are not conditioned on the requirement.

Reading rule: compare a method only where it is actually used; the baselines do not answer the same research question.

[1] §4.1.4-§4.1.6 · no truly comparable requirement-targeted baseline existed

Each requirement becomes one generation-and-test experiment.

01 · SPECIFY	02 · FILTER	03 · ADAPT	04 · VALIDATE INPUT	05 · TEST + AUDIT
Requirement + glossary Authors provide the precondition and manually encode the expected output. 25 EVALUATION REQUIREMENTS	Label original data Apply measurements, scene graphs, visual Q&A, or human labels; retain precondition matches. ONE MATCHING SUBSET	Train + prompt LoRA Pair images with a trigger phrase and precondition text, then sample candidate tests. ONE LORA PER PRECONDITION	Consistency + distributions Glossary classifiers, KID, and Jensen-Shannon divergence evaluate generated inputs. RQ1: 100 × 10 · RQ3: 1,000	Run model + oracle Check the postcondition; independently assess sampled failures for realism and precondition match. RQ4-RQ6: 1,000 × 10 · AUDIT: 330

Important: RBT4DNN assumes the requirement and glossary already exist; it does not discover them automatically.

[1] §3 and §4.1 · RQ2’s generated sample count is not clearly restated in §4.2.2

LoRA compute is measured; total workflow cost is not.

≈24.7

A100-hours if the 21 primary MNIST, CelebA-HQ, and SGSM adapters run serially.

PER PRECONDITION	LoRA training	52.1–82.5 min
REUSABLE ACROSS REQUIREMENTS	Glossary labels + classifiers	cost not reported
PER GENERATED CAMPAIGN	Sampling + model execution	cost not reported
PER AUDIT SAMPLE	Human validity review	cost not reported

DEFENSIBLE CONCLUSION

Per-precondition LoRA adds roughly linear adapter overhead; the paper cannot tell us whether compute, labeling, engineering, or human validation dominates total cost.

[1] §4.1.4 · derived total excludes ImageNet, extra LoRAs, labeling, generation, evaluation, and engineering

Six questions form a two-stage evidence argument.

Validate the generated inputs

RQ1–RQ3

RQ1 **Consistency** Does the image satisfy the precondition?

RQ2 **Plausibility** Is its distribution close to matching training images?

RQ3 **Variation** Are other glossary features distributed similarly?

Evaluate the resulting tests

RQ4–RQ6

RQ4 **Comparison** Are targeted outcomes more informative than baselines?

RQ5 **Violations** Do tests reveal credible postcondition failures?

RQ6 **Diagnosis** Do failures expose repeatable decision patterns?

Across 21 requirements, targeted LoRAs match preconditions 92.1% on average.

MEASUREMENT

One binary glossary classifier per term; mean classifier accuracy: 94.5%.

Domain means differ: CelebA-HQ automated visual-question labels 95.8%; human labels 96.1%; SGSM is weaker at 82.3%.

M3·targeted
very thick 7

69.4%
MODEL PASS

95.8%
PRECONDITION MATCH

Most outcomes address the requested scenario.

M3·all data
same prompt

99.6%
MODEL PASS

7.4%
PRECONDITION MATCH

The excellent pass rate mostly describes ordinary sevens.

MNIST Fig. 3 also includes DeepHyperion and image transformations: neither targets the requirement, and both achieve low match.

[1] §4.2.1 and Fig. 3 · glossary classifiers remain proxies, not ground truth

Targeted LoRA cuts KID by 77.09% for MNIST and SGSM; CelebA-HQ is mixed.

77.09%

average KID reduction across MNIST M1-M6 and SGSM S1-S7

CELEBA · AUTOMATED LABELS	12.9% average reduction
CELEBA · HUMAN LABELS	27% average reduction
REFERENCE	matching filtered subset

Matching training subset

KID ↓

lower = closer
distributions

Generated tests

Boundary: KID is a distribution-level proxy—not an image-level realism score—and several CelebA-HQ reference subsets contain fewer than 300 images.

[1] §4.2.2 and Table 3 · same prompts; targeted LoRA compared with prompt-only FLUX

RQ3 preserves measured feature frequencies—not unrestricted image diversity.

FILTERED TRAINING SUBSET		GENERATED TESTS		
Other glossary features		Glossary-classifier frequencies		
very thin		↔	very thin	
right leaning			right leaning	
very tall			very tall	
RESULT		NOT MEASURED		
MNIST and SGSM targeted-LoRA distributions were more than three times closer on average than prompt-only FLUX; CelebA-HQ improved about 32%.		Duplicates, memorization, pixel diversity, within-term mode collapse, and systematic latent-space coverage.		

[1] §4.2.3 and Table 4 · schematic bars explain the metric; they are not reported measurements

For M2, RBT4DNN reaches the precondition; both untargeted baselines largely do not.

GATE 1 · INPUT

Does this image actually show a very thick 3?

DeepHyperion and transformations generate recognizable digits, but they are not conditioned on M2 and rarely reach its semantic precondition.



GATE 2 · OUTPUT

Only then does the classifier outcome address M2.

RBT4DNN trades a superficially higher pass rate for much more requirement-relevant evidence.

Narrow conclusion: requirement-aware generation is a better basis for these MNIST requirements than the two tested untargeted baselines—not universally superior DNN testing.

RBT4DNN found failures for 19 of 21 requirements; reviewed false positives averaged 3.3%.

330

failing images independently assessed by multiple authors and aggregated.

3.3%

mean estimated false-positive rate across the reviewed requirements.

SAMPLE	11 requirements	30 failures reviewed for each requirement whose pass rate was below 90%.
QUALITY	All realistic	Worst estimated false-positive rate was 11.5%; assessor disagreement on precondition consistency was 1.7%.
INSIGHT	Tests expose more than model errors	S2 suggested an underspecified distance condition; S3 exposed generation and labeling difficulty.

[1] §4.3.2 and Fig. 7 · postcondition violations are observed; internal model faults are not localized

Failures reveal repeatable patterns —but not their causes.

1,486

ImageNet failures examined across the exploratory study.

3

high-accuracy models showed concentrated, model-specific predictions.

OBSERVED	Prediction clusters	Each classifier repeatedly favored particular incorrect labels for some requirements.
ASSOCIATED	Image patterns	Manual inspection found partial animals, background objects, and multi-object scenes.
BOUNDARY	Exploration, not causality	The patterns motivate hypotheses about ImageNet’s single-label supervision; they do not prove causal mechanisms.

[1] §4.3.3, Fig. 8 and Tables 6–7 · pass rates span approximately 93.5%–99.5%

The paper demonstrates feasibility—not a complete assurance process.

GENERATIVE TRAINING Better tests Improve generator training and adaptation to reveal more observed failures and reduce false positives.	COVERAGE EVIDENCE Latent coverage Adapt systematic latent-space coverage when test generation does not reveal a failure.	GENERALIZATION More domains Expand driving-like safety datasets and explore broader robustness and functional requirements.
ASSURANCE BOUNDARY Automated labels, selected datasets, narrow baselines, and small KID subsets limit how far the evidence generalizes.		

[1] §4.4 and §5 · author-stated scope kept separate from this seminar's critique

The paper jointly validates 330 sampled failures—but not the full failure population.

PRECONDITION INVALID · MODEL PASSES Irrelevant pass The output says nothing about the stated requirement.	PRECONDITION INVALID · MODEL FAILS Off-target false positive A model error is not yet a requirement counterexample.
PRECONDITION VALID · MODEL PASSES Valid requirement pass Relevant evidence, but never proof of the universal property.	PRECONDITION VALID · MODEL FAILS Credible counterexample This joint event is the paper’s strongest testing evidence.

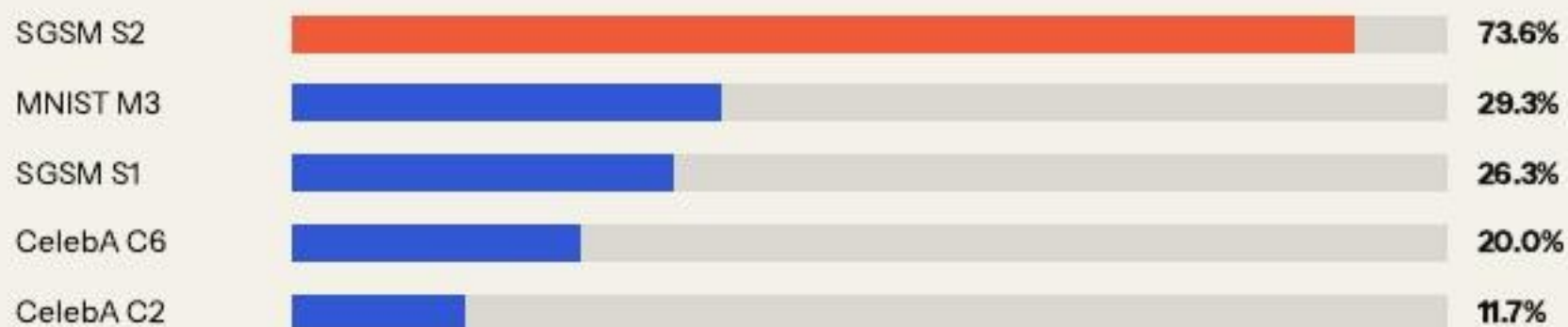
SAMPLED JOINT AUDIT

30 failures × 11 low-pass requirements = 330 independently assessed samples.

Fair critique: the paper did perform joint validation. Its generalization remains limited by author assessment, fixed samples, no reported confidence intervals, and no external or blinded annotators.

The proxy and 42-image LLM check identify audit targets; neither estimates population validity.

Independence-proxy ranking



Diagnostic only · unconditional $P(\text{match}) \times P(\text{fail})$ · not an observed joint rate

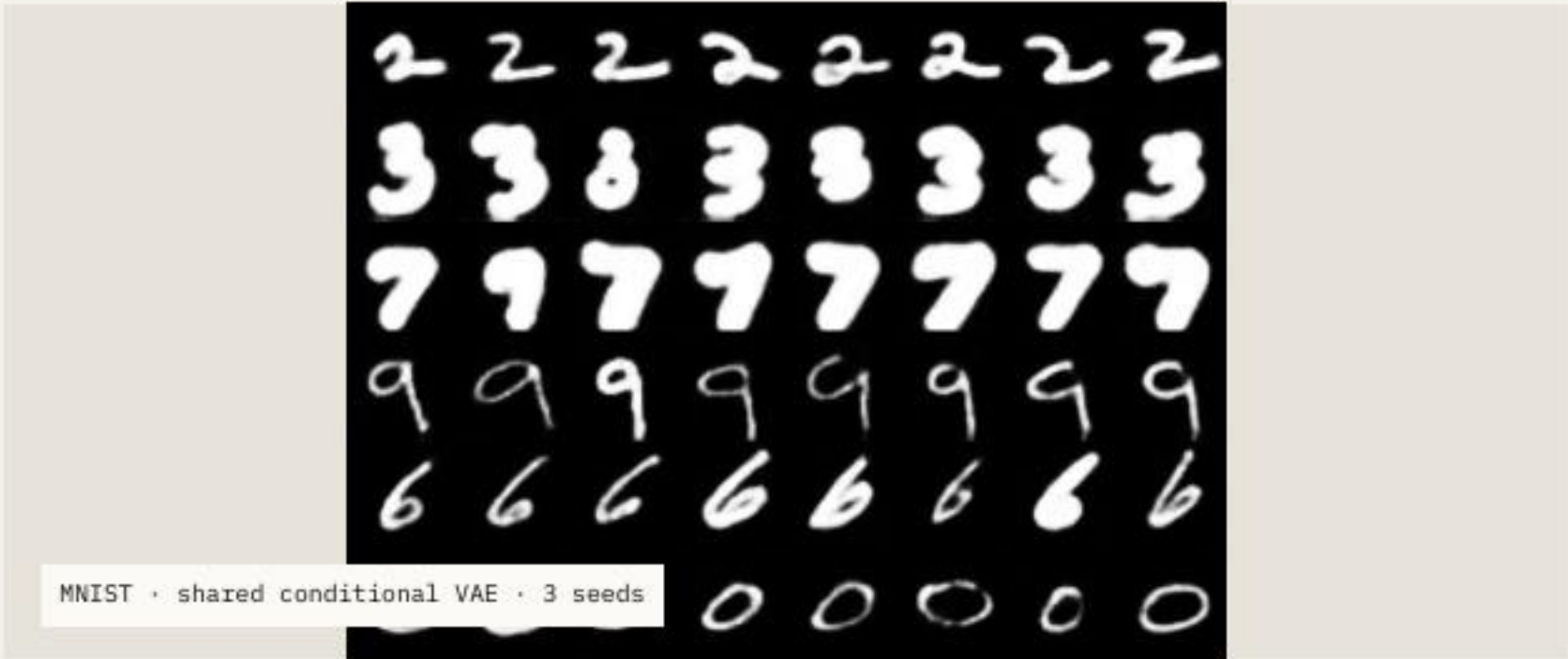
SMALL EXTERNAL SEMANTIC CHECK

.738

31 of 42 sampled images were judged precondition-valid by one multimodal model.

Use: decide where independent joint review is most valuable.

The product assumes independence. The paper's conditional human audit among failures is methodologically stronger; this LLM check is only a second opinion.



.945

mean classifier pass rate

.938

paper v3 repeated-run mean, M1-M6

The shared VAE matches MNIST digit-pass rates, but precondition validity was not measured.

Weak case	M3 performance	.743
CelebA	generated-image alignment	.613
Evaluator	CelebA evaluator accuracy	.636

Conclusion: the similar classifier pass rate is a promising pilot in a simple domain. Evaluation pipelines differ, and it says nothing yet about precondition validity, realism, diversity, FLUX + LoRA, or natural images.

Requirements make tests **relevant**. Validity makes failures **credible**.

WHAT THE PAPER ESTABLISHES

Structured requirements can guide realistic, diverse, and scenario-specific neural-network test generation across four domains.

WHAT REMAINS OPEN

Broader independent joint audits, shared generators, alternative adaptation methods, and end-to-end replication at larger scale.

Overall: a strong and original testing direction, with an incomplete assurance argument—and clear opportunities for the next study.

References.

[1]

N. J. Mozumder, F. Toledo, S. Dola, and M. B. Dwyer. “RBT4DNN: Requirements-based Testing of Neural Networks.” arXiv:2504.02737, v3, 2025.

[2]

E. J. Hu et al. “LoRA: Low-Rank Adaptation of Large Language Models.” ICLR, 2022.

[3]

RBT4DNN artifact. github.com/less-lab-uva/RBT4DNN · seminar provenance pinned to commit c3deff871ed41043d83a05289faab84c41706586.

[4]

Seminar experiments. Reproducible notebooks, results, summaries, and provenance in this repository.

Original paper ↗	Original artifact ↗
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